Arab academy for Science, Technology and Maritime Transport

Binary Addition

Rules of Binary Addition

- **0 + 0 = 0 ; Carry = 0**
- **0 + 1 = 1 ; Carry = 0**
- **1 + 1 = 0 ; Carry = 1**

- **0 + 0 + 0 = 0 ; Carry = 0**
- **0 + 0 + 1 = 1 ; Carry = 0**
- **0 + 1 + 1 = 0 ; Carry = 1**
- **1 + 1 + 1 = 1 ; Carry = 1**

Example:

1) $(1011)_2 + (111)_2$

```
  1 1 1
  1 0 1 1
+   1 1 1
---------
  1 0 0 1 0
```

2) $(110010)_2 + (111001)_2$

```
    1
  1 1 0 0 1 0
+ 1 1 1 0 0 1
-------------
  1 1 0 1 0 1 1
```

3) $(100010)_2 + (101100)_2$

```
    1 0 0 0 1 0
+   1 0 1 1 0 0
---------------
  1 0 0 1 1 1 0
```

Add the following binary numbers:

1. $(10001)_2 + (11101)_2$
2. $(10111)_2 + (110001)_2$
3. $(110110)_2 + (1111)_2$
4. $(11100)_2 + (10101)_2$
5. $(110)_2 + (1010)_2 + (1001)_2$
6. $(100010)_2 + (110111)_2$

Add the following decimal numbers using 8-bit Two's Complement binary representation:

1. 5 + (-3)
2. 7 + (-5)
3. (-2) + 5
4. 13 + (-9)
5. (-17) + 2
6. (-64) + 32

Reminder: Represent each operand and the result using exactly 8 bits. Use Two's Complement to represent negative numbers and perform addition (e.g., 5 = 00000101, -2 = 11111110).